Wesley Brown

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TECHNICAL SKILLS

Languages: Python, R, SQL, TypeScript, Julia, C++, Java, GQL
Frameworks & Tools: PyTorch, TensorFlow, Docker, Kafka, Git, Jupyter, MongoDB, Postgres, Neo4j, AuraDB, AWS S3, Tableau, Linux, Conda, Cloud Platforms
Specialties: Computer Vision, Time Series Analysis, ETL Pipelines, Geospatial Analysis, Graph Databases, Data Modeling, Database Design

Work Experience

Computational Infrastructure for Geodynamics **July 2026 - Present**
Graduate Data Engineer (Graduate Student Assistant IV) *Davis, CA*

- Architect end-to-end ETL pipelines and design network server storage layouts to integrate legacy digitized seismogram records into high-performance computing frameworks.
- Design and optimize a Neo4j graph storage solution from scratch, applying database design and data modeling to automated parsing pipelines that efficiently structure metadata into AuraDB nodes and relationships.
- Optimize open-source geodynamic software through code refactoring, version control management, and containerized deployment protocols.

Professor Can Le, UC Davis **July 2025 - September 2025**
Undergraduate Student Researcher *Davis, CA*

- Investigated the efficiency and bias of cross-validation-tuned estimators for combining multi-source data, improving overall model robustness.
- Conducted over 40,000 simulations to rigorously benchmark estimator performance against theoretical optimal standards.

Real AI Dynamics **April 2024 - July 2024**
Technology Development Intern *Remote*

- Developed two micro-SaaS AI tools from conception to deployment, enhancing platform functionality and user experience.

PROJECTS

IoT Generation & ETL Pipeline

- Python, Polars, PostgreSQL, AWS S3, Docker, Grafana
- Developed an automated end-to-end ETL pipeline simulating connected vehicle telemetry, using AWS S3 for scalable cloud storage and PostgreSQL for structured data management.
- Implemented efficient data transformations with Polars and loaded structured records into PostgreSQL, using Docker Compose for containerization and Grafana for pipeline output visualization.

Air Quality Spatial Modeling

- Python, PyTorch, Time Series, Geospatial Analysis
- Led a team of five in the Davis Data Science Club to analyze local air quality using environmental and demographic data, applying analytical and problem-solving skills to identify community risks.
- Utilized time series forecasting and geospatial mapping to identify high-risk communities and deliver actionable insights for targeted protective measures.

Education

Master of Science, Analytics (Computational Track) **August 2026 - May 2028**
Georgia Institute of Technology

- **Coursework:** Analytics Modeling, Foundations of Business

Bachelor of Science, Data Science **September 2024 - June 2026**
University of California-Davis

- **GPA:** 3.79
- **Coursework:** Applied Machine Learning, Applied Time Series Analysis, Data Processing Pipelines, Probability Theory